NOTICE 1
23 October 2012

TEST OPERATIONS PROCEDURE

This Test Operations Procedure (TOP) has been archived per ATEC memorandum dated 18 October 2012.

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13. ABSTRACT

This document provides procedures and techniques for testing the ability of chmical munitions, weapons and equipment to withstand exposure to and to function within a hot humid tropic environment. Equipment includes munitions, dissemination, decontamination and detection device and individual and collective protective equipment. Test Description covers initial inspection of equipment, operational performance, storage aspects, surveillance, maintenance, safety, human factors and value analysis.

DD FORM 1473 REPLACES DD FORM 1473, JAN 65, WHICH IS

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REFERENCE FILE

Materiel Test Procedure 8-4-003(M)
13 October 1970 Tropic Test Center

US ARMY TEST AND EVALUATION COMMAND
COMMON TROPIC ENVIRONMENTAL TEST PROCEDURE

CHEMICAL EQUIPMENT

1. OBJECTIVE

This document provides test methodology and testing techniques for determining the capability of chemical munitions, weapons and equipment to withstand exposure to, and function effectively within, humid tropic environments. A principal purpose of tropic testing is to compare data obtained before and after dynamic tests, and before and after static storage tests.

2. BACKGROUND

a. Army worldwide operations create a need for developing and testing new materiel. This creates a need for determining whether chemical munitions, weapons and equipment, and their accessories (hereinafter called test items) will meet specified performance requirements when they are exposed to natural wet-warm or wet-hot tropic environment conditions. Testing under these conditions is generally not conducted until data from simulated environmental tests provide reasonable assurance that the test item will function satisfactorily when subjected to such tropic environmental conditions.

b. See MTP 2-4-003, Appendix A for details regarding the environmental characteristics of those areas in the Panama Canal Zone, and in the Rio Hato Training Area, which are available for testing, major problems associated with the tropic areas, a description of fungus-inert and fungus-susceptible materials, the corrosion-resistant properties of metals and the major environmental effects caused by exposure to the Panama Canal Zone climate.

c. Chemical equipment either disseminates, protects against, detects and identifies, or neutralizes or decontaminates chemical agents. However, also involved in this category are items employing flames, smoke and incendiary agents which are usually identified with conventional weapons.

Chemical items can be grouped together as follows, according to the principal functions they perform:

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REFERENCE FILE

1) Munitions

These items are designed to deliver chemical and riot control agents, smoke, or incendiaries to selected targets.

2) Individual Protective Equipment

These items are designed to protect the individual from the effects of chemical agents by: (1) preventing the entry of such agents into the respiratory system, or (2) decreasing the cutaneous and percutaneous hazards.

3) Dissemination Devices

This equipment is intended for use in distributing relatively large quantities of chemical and riot control agents or smoke over a large area. Such items are designed to disseminate bulk, micro-pulverized powders or liquid agents.

4) Decontamination Devices

These items are designed to facilitate the process of making any person, object, or area safe by absorbing destroying, neutralizing, making harmless, or removing chemical agents.

5) Detection Devices

This equipment is intended to provide friendly troops with sufficient notice of a chemical agent attack to enable them to take adequate protective measures. Also included in this category are items with the capability of sampling and identifying chemical agents, detection of personnel, or marking and tracking of personnel.

6) Collective Protective Equipment

These items of equipment are designed to provide and maintain filtered air under positive pressure at a level which precludes the infiltration of hazardous levels of toxic agents. They are adaptable for use with field medical facilities, and other shelters such as command posts, communications centers, portable pressurized shelters, and certain vehicles and vans.

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d. Test Site - The test site must be selected with care. It must be sufficiently isolated to prevent inadvertent exposure to chemical agents, and be of adequate dimensions to allow proper evaluation of the dispersion characteristics if a munition or dissemination device is being tested. The type of terrain, vegetation, and prevailing atmospheric conditions are factors to be considered in test site selection.

e. Special Considerations (Agents and Agent Dissemination) - Instrumentation is an essential part of testing. Because chemical agents, once released, are wholly dependent upon the prevailing atmospheric conditions and those surface effects produced by terrain features and vegetation in the target area, adequate measuring instruments to record temperature, temperature gradient, wind speed and direction, atmospheric pressure, and relative humidity must be employed. Once an agent is released, it is imperative to determine the dosage at various locations in the horizontal and, if possible, vertical plane. The proper selection of sampling devices to collect these agents if of utmost importance.

f. Special Considerations (Protective Equipment) - In testing individual protective equipment, human engineering aspects are of paramount concern. Testing protective masks, for instance, involves not only determining if they will deliver chemical agent-free air to the wearer, but also evaluating the following factors: (1) comfort of fit while wearing, (2) ease of breathing (inhalation and exhalation), (3) visibility (with and without optical inserts), (4) interference with other equipment carried or used, (5) interference with normal duties, (6) ease of donning, (7) comfort during exposure to high temperature and humidity, (8) the nosecup fit, (9) adequacy of speech transmission, and (10) mask stability. Data pertaining to the technical characteristics are gathered as a result of laboratory testing; human factors aspects are best arrived at by the interrogation of test subjects who participated in a series of wearing tests. This questioning should be conducted by a human factors engineer whenever possible.

Protective clothing must also be subjected to a human factors evaluation. It must be confortable to wear, not cause skin irritation, produce a minimum of heat fatigue, and be easily donned or doffed. In testing protective clothing, a human factors engineer should be consulted to devise a suitable method of questioning wearers concerning these aspects.

Collective protection provides chemical agent-free air to a number of individuals who would otherwise be hampered by wearing a protective mask. The unit is connected to a shelter, a vehicle, a van, or a fixed installation and causes a positive pressure to be built up therein, which precludes infiltration or chemical agents. It must be rugged, dependable, and compatible with the equipment it serves.

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g. Special Considerations (detection and Surveillance Equipment) - The value of this type of chemical equipment lies in its sensitivity, quick response time, and low false alarm rate. In testing, particular attention must be paid to these characteristics in order to derive valid data. Human factors are involved: (1) in testing the audibility or visibility of an alarm signal, and (2) in determining if the operator can successfully manipulate a detector kit and properly recognize the results.

h. Special Considerations (Decontamination Equipment) - Caution must be exercised in the use of decontamination agents to avoid the irritation to eyes, skin, and respiratory tract which some chemicals produce. The reaction of some decontamination agents with certain chemical agents resulting in toxic fumes or flashing should be avoided or minimized.

3. REQUIRED EQUIPMENT

One or more of the following items and/or facilities may be required to obtain data during the various evaluations:

a. Facilities and measuring and recording devices listed under paragraph 3. "REQUIRED EQUIPMENT" in MTP 2-4-003, as applicable.

b. Reference or comparison items, as required.

c. Facilities and measuring and recording devices as listed in the applicable commodity item MTP.

d. Non-toxic agent simulant, RCA (Riot Control Agent), and associated containers and dispersers.

4. REFERENCES

A. AR 11-33 Chemical and Biological Surety Program.
B. AR 50-21 Chemical and Biological Accident and Incident Control (CBAIC).
C. AR 70-38 Research and Development: Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions.
D. USAMC Regulation 11-33 AMC Chemical and Biological Weapons Surety Program.
E. USAMC Supplement 1 to AR 50-21 AMC Chemical and Biological Accident and Incident Control (CBAIC).
F. USAMC Regulation 385-12 Safety: Verification of Safety of Materiel from Development Through Testing and Supply Disposition.
G. USAMC Regulation 385-100 Safety Manual (dated Apr 70).
H. USAMC Regulation 385-102 Safety Regulation for Chemical

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Agents, GB and VX.

I. USAMC Regulation 385-235 Safety: Storage and Handling Criteria for Chemical Ammunition, Explosives, Chemical Warfare Agents, and Hazardous Chemicals.

J. USAMC Pamphlet 702-3 Quality Assurance Reliability Handbook.

K. USAMC Pamphlet 706-134 Engineering Design Handbook: Maintainability Guide for Design.

L. USATECOM Regulation 11-33 USATECOM Chemical and Biological Weapon Surety Program.

M. USATECOM Regulation 70-23 Research and Development: Equipment Performance Reports (EPRs).

N. USATECOM Regulation 385-6 Safety: Verification of Safety of Materiel During Testing.

O. USATECOM Regulation 700-1 Quality Assurance: Value Engineering.

P. USATECOM Regulation 750-15 Maintenance of Supplies and Equipment: Maintenance Evaluation During Testing.

Q. USAGETA Document Human Factors Evaluation Data for General Equipment (HEDGE).

R. MIL-HDBK-472 Maintainability Predictions.

S. MIL-STD-794 Parts and Equipment, Procedures for Packaging and Packing of.

T. MIL-STD-1472 Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

U. MTP 2-4-003 Tropic Environmental Test. Wheeled, Tracked and General Purpose Vehicles.

V. MTP 3-4-003 Tropic Environmental Test. Armament and Individual Weapons.

W. MTP 4-4-003 Tropic Environmental Test. Ammunition and Explosives.

X. MTP 4-3-520 Field Storage.

Y. MTP 8-4-004 Long Term Surveillance/Environmental Testing of CBR Munitions, Weapons and Equipment.

Z. MTP 8-2-500 Receipt Inspection.

AA. MTP 8-3-500 Physical Characteristics.

AB. MTP 8-2-503 Rough Handling and Surface Transport.

AC. MTP 8-3-506 Safety Hazards.

AD. MTP 8-3-507 Maintenance.

AE. MTP 8-3-509 Human Factors Engineering.

AF. MTP 8-2-510 Decontamination.

AG. MTP 8-2-511 Leak Testing of Protective Equipment.

AH. MTP 8-2-512 Leak Testing of Agent-Filled Munitions and Containers.

AI. MTP 8-2-513 Dissemination Characteristics, CB Munitions/Dissemination Devices.

AJ. FM 31-30 Jungle Training and Operations.

AK. Specific Volume 8 MTP for commodity item under evaluation.

AL. Annex C to USATECOM Disaster Control Plan - USATECOM Chemical and Biological Accident and Incident Control (CBAIC).

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5. SCOPE

This MTP is intended to be used as a guide in conjunction with the MTP for the specific commodity item(s) undergoing service tests. Reference is made to other MTP's for the actual test procedures to be followed in assessing test item physical condition, technical characteristics, and operating performance. The primary intention of this MTP is to specify the environmental conditions under which such testing is to be performed, collect data relating test item performance to continued tropic exposure, and evaluate the data collected to determine the suitability of the test item for use in the tropics.

5.1 SUMMARY

This MTP describes the following tests to be conducted on chemical equipment and their accessories.

5.1.1 Preparation for Test

This section provides guidance for test project planning, including a discussion of facilities, documentation, calibration, test scheduling, and equipment required, test personnel training and familiarization, statistical planning, and the preparation of an appropriate mission scenario.

5.1.2 Test Conduct

a. Initial Inspection and Operation - A determination of the condition of the test item upon its arrival at the installation site, and an assessment of its performance characteristics before it has been exposed for any appreciable length of time to the tropic environment in which it will be used.

b. Operational Performance - An evaluation to determine the capability of the test item to perform its intended mission. This test is conducted during both dry and rainy seasons when the test schedule permits.

c. Short Term Storage - A determination of any deterioration in the test item occurring during and after short term storage.

d. Surveillance (Long Term Storage) - A determination of any deterioration in the test item occurring during and after long term storage, when properly packed and packaged for storage in a tropic environment.

e. Maintenance Evaluation - An evaluation to determine the maintenance characteristics of the test item and the suitability of the maintenance test package in a humid tropic environment.

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f. Safety - An evaluation to determine the safety characteristics and possible hazards to which the test item is exposed.

g. Human Factors - An evaluation to determine those design and performance characteristics affecting the test item user which are peculiar to the tropic environment.

h. Security from Detection - An evaluation to determine the capability of the test item to avoid detection when stored or operated in a humid tropic environment.

i. Value Analysis - An evaluation directed at analyzing the primary function and features of the test item for the purpose of cost reduction without compromising performance, reliability, quality, maintainability or safety.

5.2 LIMITATIONS

a. The materiel presented in this MTP is limited to field testing. Guidance for testing equipment in simulated environments or other induced wet-warm or wet-hot conditions has been intentionally omitted.

b. Procedures outlined in this MTP do not constitute detailed test plans. A test plan for tropic environmental testing of an item may be prepared using the guidance given in this MTP, but each test activity preparing environmental test plans must make its own judgment as to the applicability of procedures, and must determine how best to obtain the required data from each item under test.

c. These procedures are limited to tests on chemical munitions, weapons and equipment which require exposure to those combinations of weather and terrain conditions, described in Appendix A of MTP 2-4-003, which are available in the Panama Canal Zone or in the Rio Hato Training Area. It is assumed that these test items have been subjected to, and successfully passed, those tests which are unaffected by ambient weather or terrain conditions, or which can be appropriately conducted in a temperate climate to determine if the test item meets the criteria. Performance tests are also limited to tests using non-toxic chemicals.

d. Toxic chemicals will not be disseminated.

e. To reduce the number of variable to a minimum, all test items must be chosen from the same production lot, and they must be also identified, when more than one test item is involved.

6. PROCEDURES

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6.1 PREPARATION FOR TEST

NOTE: The quantity of test items received for test should be greater than the quantity specified in the statistical plan, except when it is known that the shipment contains no non-repairable items in which case, adequate quantity of repair parts or subassemblies should be included with the maintenance package.

6.1.1 Test Project Planning

The test project officer and other designated personnel must:

a. Conduct a thorough study of stated requirements as contained in QMRs, SDRs, TCs and the test directive to insure that complete and suitable test criteria are selected.

b. Review all instructional material issued with the test item by the manufacturer and reports of previous conducted on similar equipment.

c. Prepare a detailed test schedule showing proposed time periods allotted for each test. Insure that cross-country testing is scheduled for test courses with both cleared and uncleared terrain in wet-warm and wet-hot environments. These environments and test courses are defined in paragraph 3 and Appendix A of MTP 2-4-003. Testing should be scheduled to take place in the peak of the wet season (October through November) and in the dry season (January through March). Wherever applicable, testing should also be conducted in areas that expose the test item to salt-laden air and salt-spray.

d. Prepare record forms and test logs for systematic entry of data, chronology of test, and evaluation in the final analysis of the test item's suitability for use in the humid tropics.

e. Review the safety release. Tabulate the safety precautions that must be followed to insure safe operation of the test item(s); and include all applicable safety Standing Operating Procedures (SOPs).

f. Verify that test facilities, equipment, and accessories are available, operational and have been properly calibrated within the time limit. Power sources, when used, shall be checked to insure correct outputs.

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g. Final arrangements for supporting or participating agencies, activities and facilities shall be made.

h. Operating instructions for test instruments to be used during test conduct should be obtained and made available to test personnel.

i. Short term storage should last not less than 6 months, and should include both dry and wet seasons. It is desirable to start all testing during the dry season, because best performance is usually obtained at that time, and this performance can then be used as a standard against which wet season performance can be compared.

j. Plan the use of photographic techniques where necessary to record and document test results.

6.1.2 Test Personnel Training and Familiarization

a. Evaluate the adequacy of all draft technical manuals and safety instructions used by the test personnel. This evaluation shall continue throughout the course of the test.

b. Instruct personnel as follows:

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Table I. Personnel Instructions

Personnel	To be instructed in
All	a. The purpose of the test. b. The characteristics of the test item. c. The characteristics of the reference item, if any. d. The kind of data to be obtained. e. The terrain(s) in which testing is to be conducted. f. The health precautions to be observed in the terrain(s) required for test conduct.
Test and reference item operators	a. The test item (and reference item, when used) operating characteristics and limitations. b. The expected test item performance. c. The safety precautions to be observed. d. The kind and extent of all maintenance actions to be taken under all specified environmental conditions. e. Operating procedures to be followed under all environmental and terrain conditions. f. The procedures to be followed when filling out checklists and questionnaires, and their purpose.
Test evaluators	a. The purpose and use of all checklists and questionnaires, and the methods to be used in their evaluation. b. The calculations to be made in evaluating test data.
Topographical analysis support	a. Physical terrain features to be measured, including water velocity and terrain slopes. b. Vegetation features to be noted and described.

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Table I. (Cont'd)

Maintenance support	a. The schedule maintenance requirements to be met. b. The procedures, equipment and material to be used to make emergency repairs and unscheduled maintenance. c. The kind of recovery vehicle to be used. d. The recovery procedures to be used. e. The maintenance records to be written.
Meteorological support (TTC Reg. 705-4)	a. The terrain areas which will be used. b. The calendar dates of all tests. c. The readings to be taken.
Storage and surveillance test	a. Storage, handling, layout, spacing, pest control, fire protection, security, and inspection procedures to be followed. b. Required organizational maintenance to be performed, including cleaning, and microbiological inspection. c. The calendar dates of all inspections.
Explosive ordnance disposal personnel	a. The terrain areas which will be used. b. The decontamination procedures to be followed.

c. When a reference item is used for comparison purposes, subject the reference items to procedures described in steps a. and b.

6.1.3 Statistical Plan

6.1.3.1 General

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a. Unless it is known, or unless it can be reasonably assumed, that the time to failure of the test item follows an exponetial or other known distribution, cross-country mission reliability should be expressed as a probability of success, using the "attributes" approach (binomial distribution), wherein the mission (or "experiment"), is satisfactorily completed, is called a success, and if the mission (or "experiment") is not satisfactorily completed, it is called a failure.

b. When the "attributes" approach is used, the preciseness of the probability figure obtained increases as the number of "experiments" increases. For a probability of success of 10%, at a lower 90% confidence level, is the best that can be claimed when the "experiment" is repeated with the same, or another identical, test item. If, however, five identical "experiments" are successfully completed, the probability of success becomes 63.1% at a lower 90% confidence level when another identical "experiment" is made. Therefore, when the "attributes", or missions (NxT) should be made as large as possible. The decision regarding the number of test items, N, to use, and the number of identical experiments, T, to make, should form an essential part of every statistical plan.

c. To obtain the maximum amount of useful information, use the same test personnel in both the wet season and in the dry season.

d. When a reference item is used for comparison purposes, follow the same test procedures as those followed with the test item.

e. Follow the instructions in AMCP 702-3 when calculating test item reliability and unless otherwise specified, use a lower confidence level of 90% in these calculations.

f. The total number of test items to be used depends on a large extent upon its "wear" or "use" characteristics. For example, a projectile, grenade, or rocket cannot be restored to its initial condition after it has been "fired". Such items therefore require the use of a new test item from the same production lot when repeated tests are to be made. Other items, such as repairable chemical agent detector kits and mechanical smoke generators, can be operated and repaired many times without any appreciable change in their initial, or "new" condition. Between these two extremes, there exists many gradations in the "wear" or "use" characteristics of different test items.

When a "one-shot" device such as a chemical land mine is being tested, the items will be divided into equal groups and a cyclic schedule designated. This cyclic schedule should encompass the various changes in meteorological conditions, See the following example:

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Table II. Cyclic Test Schedule

Cycle	Test to be Performed
No. 1	Initial Performance
No. 2 (3 months)	Dry Season/Wet Season Performance
No. 3 (6 months)	Short Term Storage Performance
No. 4 (9 months)	Wet Season/Dry Season Performance
No. 5 (12 or XX months)	End of Test Performance/or Long Term Storage (Surveillance)

When a "long-wearing" device, such as a portable decontamination apparatus, is being tested, and which can be repaired many times without any appreciable change in performance characteristics, one group of test items from the same production lot is all that is required, provided that the item is restored to its "as-new" condition at the start of each of the environmental tests described above. A similar approach can be used without too much loss in accuracy, when testing "medium-wear" devices such as protective clothing or shoes.

6.1.3.2 Chemical Munitions

Establish a statistical plan in accordance with the procedures contained in paragraph 6.1.4 of MTP 4-4-003, including the division of the lot into five test groups.

6.1.3.3 Individual Protective Equipment

Establish a statistical plan in accordance with the procedures contained in paragraph 6.1.3.2 of MTP 3-4-003 except substitute the word "wear" in place of the word "fire", and the word "device" in place of the word "weapon".

6.1.3.4 Dissemination, Decontamination and Detection Devices

Establish a statistical plan in accordance with the procedures contained in paragraph 6.1.3.2 of MTP 3-4-003 with the following exceptions:

a. Substitute the word "operate" in place of the word "fire".

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b. Substitute the word "device" in place of the word "weapon".

6.1.4 User Test

a. The user test will describe the tests to be conducted under simulated field conditions by personnel who would use the equipment in the field. The mission scenario (ref. Appendix C) will describe the test areas and operations that will be used for evaluating the use of the test item.

b. Unit missions, where possible, will include both defensive and offensive operations as they apply to the type of item being tested, i.e., an individual hand-held item will be evaluated as a defensive weapon for a battalion perimeter, and then as a part of an attack force securing an enemy position.

c. The mission scenario should include, as a minimum, the following:

1) Details of test item prime function(s) to be performed in each condition of environment and terrain.
2) Sketch map, or photomap, of the test area(s), and a description of the environment(s) to be found therein.
3) Number of test items and personnel.
4) Actions or movements to be used.
5) Number of support equipment and personnel.

d. When test items are evaluated during tactical training exercises, the test item use will be documented from the operation orders, scenarios used in conjunction with training problems, situation outlines and Army Training Tests.

6.2 TEST CONDUCT

NOTES: 1. Insure that a safety release statement has been received from USATECOM before making any tests.

2. Observe all specified safety requirements at all times.

3. Performance assesment shall be accomplished by observers equipped with the means of recording visual, aural, and judgmental observations and related time factors. Observer activities shall not interfere with, or influence in any manner, the

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functions of the test item or operators. Subtest shall be conducted concurrently or in conjunction with other subtests whenever possible, so that the time taken to collect the required data will be minimized. Testing shall be conducted under all meteorological conditions prevailing during the test period.

4. When the test item meets specified operational tests during the course of an assigned mission, the test item performance is defined as being a mission success.

6.2.1 General Requirements

a. Conduct all tests under prevailing weather conditions, except that wind speed and direction requirements are to be met when performance tests are being made on the dispersion characteristics of smoke or chemical agent "clouds".

b. When a reference or comparison item is used, subject it to the same tests as those applied to the test item.

c. When the test item will be exposed to the shock, vibration, vegetation, terrain and weather conditions encountered during a cross-country mission, observe the following:

1) Travel over the test course during daytime and nighttime unless safety requires only daytime testing.
2) Travel over grass lands during both dry and wet seasons.
3) Ascertain that the test course includes the terrain and vegetation conditions under which the test item was designed to be operated.
4) When a reference or comparison item is to be carried cross country, take it over a course parallel to that taken by the test item, and trail it behind the test item a sufficient distance to avoid the danger of it becoming damaged by the same terrain feature which could damage the test item.

d. Note the ambient temperature, temperature gradient (½ and 2 meter level), rain, cloud cover, relative humidity, wind speed and direction for each test.

e. Conduct all initial inspections, maintenance actions,

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and operations as specified in the applicable draft technical manuals.

f. At the end of each operational test, ask the test personnel to fill out a questionnaire giving their opinion regarding the performance of their own test item, and its relative merits or demerits.

NOTE: All questionnaires will be reviewed by human factors personnel.

g. Subject comparison items to the same tests as those applied to the test item.

h. Safety Statement or Safety Release

The test officer shall insure that a safety statement or safety release pertaining to the item undergoing commodity engineering or service test has been received from the developing agency and is understood before the test is started. The safety statement includes information pertaining to operational limitations and specific hazards peculiar to the test item.

i. Safety

1) Test and subtest plans and procedures shall be prepared to insure performance in the safest manner consistent with conduct of the USATECOM evaluation. The cardinal principle is to limit exposure of a minimum of personnel, for a minimum time, to a minimum amount of hazardous material consistent with safe and efficient operations. Plans shall include safety procedures, precautions, protection, and emergency procedures, as necessary. Technical information on the hazards and safety characteristics of the test item, as provided by the Safety Statement and other pertinent information, shall be included. Such information shall include evaluation of potential hazards, analysis of risks, limitations, and precautions including special test equipment and techniques that should be incorporated in test plans and procedures.
2) A specific individual shall be charged with responsibility for safety. He shall be familiar with the construction and operation of the test item and its critical components, shall have full knowledge of the hazards and safety aspects of the test, and shall review test procedures for evaluation of hazards and recommend control measures.

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3) All personnel who participate in or observe the tests shall be briefed on the hazards involved and the precautions required to insure proper test methods and procedures.
4) Observe all of the safety precautions specified in AMCR 385-224, AMCR 385-100, MTP 8-3-506, the applicable Volume 8 commodity item MTP and the draft manual.
5) For safety, protective masks and appropriate protective clothing shall be worn by testing personnel when suspected or actual leaks are present in agent-filled munitions and containers.
6) Any test ammunition or explosive considered to be unsafe shall be disposed of by explosive ordnance disposal (EOD) personnel.

6.2.2 Initial Inspection

Perform the following immediately upon receipt of the test item:

a. Inspect and record the condition and identification marking of all packages and packing and their conformance to MIL-STD-794 for overseas shipment for Level A or Level B, as applicable.

b. Weigh and measure the individual package(s) for each packaged test item component, or a sample package, if identical components are contained in several packages. Unless otherwise specified, the minimum sample size shall be as shown in Table III.

c. Unpack and remove all preservative coatings on test item(s).

d. Verify completeness of the test item, associated components, and maintenance test package as specified in the Basic Issue Item List (BIIL), and submit an Equipment Performance Report for any damages or omissions, if required.

e. Determine the physical characteristics of the test item(s) in accordance with the appropriate procedures in the applicable Volume 8 MTP and in MTP 8-3-500.

f. Inspect all surfaces for proper painting and other protective coatings. Take color photographs of all surfaces displaying roughness, corrosion or microbiological growths. When possible, include a calibrated color patch standard.

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g. Conduct an operator preoperational inspection in accordance with the draft technical manual, MTP 8-2-500 (receipt inspection) and MTP 8-2-512 (leak testing), as applicable, and repair or replace all test items or test item components which do not pass this inspection.

Table III. Minimum Sample Size

Number of Items	Minimum Sample Size
1-4	all
5	4
6	4
7	5
8	5
9	6
10	6
11-15	7
16-40	8
41-80	9
81-1000	10
1000 or more	1 %

6.2.3 Initial Operation

a. Assemble and/or emplace or position the test item in the physical environment in which it will be used.

b. Operate or exercise the test item as described in the applicable Volume 8 MTP, (including MTP 8-2-511 when leak testing

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protective equipment) in all specified modes and ranges of use, under the prevailing meteorological conditions, and determine its effectiveness.

NOTE: It is preferable to perform initial operation tests during the January-February-March dry season because best performance is most likely to occur then. This will then provide a performance standard against which all other operational tests can be compared.

6.2.4 Operational Test

Perform tests during the dry season, which normally starts in January and ends in mid-April and during the rainy season, which normally starts in mid-April and ends in December.

a. Perform the tests described in paragraph 6.2.3 b. for a sufficient number of missions, to obtain a suitably precise measure of the test item's probability of mission success. (See paragraph 6.1.3.) When specified, these tests shall include subjecting the test item to the portions of MTP 8-2-503, including the shock, vibration, weather, water immersion and vegetation conditions encountered during cross country, secondary road, or airborne operations, as applicable.

b. During mission operations, verify that no terrain conditions or obstacles exceed the limits allowed for the test course, and measure or estimate the water velocities and depths at all stream and river crossings.

c. When testing small chemical munition containers, determine the "skittering" effect caused by the release of the agent (when criteria requires skittering of the munition).

d. Dissemination characteristics shall be determined as described in MTP 8-2-513.

e. Decontamination procedures shall conform to MTP 8-2-510.

6.2.5 Short Term Storage

Storage of the test item and/or components shall be consistent with normal practice for the commodity class of the test item as supplemented by the applicable information in the draft technical manual.

a. Place the test item and/or components in the area designated for short term storage. Unless otherwise specified, minimum use of preservatives should be made.

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b. Periodically inspect the test item for evidence of corrosion or microbiological growths. Sample and identify any growths detected. Unless otherwise specified, make these tests only on a sample quantity of test items as specified in Table III. When chemical munitions are being tested, break open one sample container during this inspection and determine if there has been any chemical reaction during the storage interval involved.

c. At the completion of the storage period, repeat the procedures of paragraph 6.2.3 to determine if any deleterious effects have been induced in the test item by storage.

d. Observe the storage requirements for ammunition outlined in MTP 4-3-520.

6.2.6 Surveillance (Long Term Storage) Test

NOTE: Unless otherwise specified, the requirements of MTP 8-3-004 shall be met throughout this test.

Perform and record the following:

a. Unless otherwise specified, store all items in their normal long term storage containers as specified in the draft technical manual.

b. Determine the kind of inspections to be made on the test items during the total storage period. Unless otherwise specified, the number of inspections to be made shall be eight, and the "cycle number" dates when these occur shall be as specified in Figure 1, MTP 8-4-004.

c. Determine at which "cycle number" dates (month, day and hour) the specified number of test items are to be operated or exercised. Unless otherwise specified, the number of test items to be operated or exercised at each specified "cycle number" operating date shall be not less than the value of N determined as described under paragraph 6.1.3.1.

d. Store the test items in the specified environment, using the applicable procedure described in TM 743-200, unless otherwise specified. Note the particular half month, and the day and hour in which this storage takes place and denote this as the "cycle No. 1" date.

e. At each of the "cycle number" dates, perform the following on each test item in the test group:

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1) Visually examine it, and take color photographs of all corrosion and fungus growth areas.
2) Make a microbiological analysis of all materials which display evidence of microbiological attack.
3) Perform all specified operator maintenance and repairs needed.
4) When an operational test is to be made, transport the specified number of test items (see subparagraph c. above) to the operating site, perform the operational test described in paragraph 6.2.3 and determine to what extent the operating characteristics of the test items have deteriorated during the storage period.
5) When chemical munitions are being tested, break open one sample container at each inspection period and determine if there has been any chemical reaction during the storage interval involved.

6.2.7 Maintenance Evaluation

NOTE: Perform the following on those test items for which operator maintenance is prescribed.

Data expressing system maintenance factors are collected in preparing for and completing active testing. If at the completion of testing very little maintenance has been performed, representative samples of the maintenance procedures in the applicable Volume 8 MTP, MTP 8-3-507 and the draft technical manual should be completed to determine the ease in performing these actions with emphasis on the following:

1) Organizational (O), Direct Support (F), and General Support (H) maintenance requirements.
2) Operator through General Support Maintenance Literature.
3) Repair parts.
4) Tools.
5) Test and handling equipment.
6) Calibration and maintenance facilities.
7) Personnel skill requirements.

6.2.8 Safety

a. Do not begin testing until a safety release is received from USATECOM.

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b. Observe the proper safety precautions during testing, and record any conditions that might present a safety hazard, the cause of the hazard, and the steps which were taken to alleviate the hazard.

c. Observe and record all of the applicable precautions specified in the following documents.

1) The test item maintenance packages.
2) AMCR 385-224.
3) AMCR 385-235.
4) FM 31-30.
5) MTP 8-3-506.
6) Applicable portions of the specific commodity item Volume 8 MTP.
7) Applicable portions of Appendix A (Questionnaires).
8) Applicable portions of Appendix B (Checklists).

d. After completing all testing, prepare a safety confirmation (service testing) for inclusion in the final report if the test item was determined safe for use. If the test item was found to be unsafe, a detailed explanation should be provided.

6.2.9 Human Factors Evaluation

Develop task/item checklists reflecting the human factors design considerations for the test item. These checklists will allow test supervisory personnel to compare test item features against appropriate criteria and record comments to evaluate the suitability of the man item interface with particular emphasis on operations under adverse weather conditions. Detailed criteria and human factors considerations for each task may be derived from USAGETA document "Human Factors Evaluation Data for General Equipment (HEDGE) Guidebook Supplement". Consult Appendix B, MTP 8-3-509 and MIL-STD-1472 for guidance in preparing a suitable checklist.

6.2.10 Security from Detection

a. Position test item in designated test site.

b. Use natural material, nets, tarpaulins, and any other prescribed camouflage material to conceal test item. Place material about site as required to enhance concealment qualities.

c. Make ground and aerial observations of concealed item at various times of the day and at various distances from the item.

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d. Repeat foregoing steps in each test site of interest.

6.2.11 Value Analysis

a. During the conduct of all tests, test personnel shall evaluate the test item from a value versus cost standpoint. Record all pertinent comments concerning features or components which can be eliminated or modified to accomplish cost reduction without impairment of performance, reliability, quality, maintainability, or safety. The applicable portions of USATECOM Regulation 700-1 shall be used for evaluation.

b. Consideration shall be given to the topics listed below. Record appropriate comments for each topic.

1) Mission Capacity

The test item should be capable of accomplishing the specified task with only a reasonable margin of excess capability. Excess capacity and unused capability normally result in unnecessary bulk, excessive weight and unwarranted costs.

2) Simplicity

Unnecessarily complex components and systems, redundancy, and the use of unneeded parts will increase costs and maintenance efforts.

3) State of the Art

In many instances the use of recently developed, currently available, components and automated features will result in an overall product improvement and cost savings.

4) Standardization

The use of identical parts and parts currently in the military system will reduce the overall logistics burden.

5) Materials and Methods of Construction

Polished surfaces, overdone finishes, and the use of expensive materials will result in unnecessary costs if used inappropriately.

6) Tolerances

Inadequate tolerances will result in difficulties and delays in accomplishing post arrival assembly, routine

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maintenance, servicing and repair.

6.3 TEST DATA

6.3.1 General Requirements

a. When measuring attributes, which are subject to small deviations, make at least two, and preferably four, different measurements under identical test conditions, and record each measurement, as well as the arithmetic mean of these measurements. Also, record the percentage accuracy of the measuring device used.

b. Indicate the accuracy of the measuring device employed when recording measurements which must be made accurately and also record the calibration date when applicable.

c. When progressive degradation is observed on any part, describe and/or photograph the degradation, and show the "before" and "after" condition together when recording.

d. When applicable, show terrain profile and vegetation and density description in the test operation log.

e. When two or more persons are asked to fill out a questionnaire giving their opinion regarding specified features in a particular test, show the scoring values used, and the scoring results obtained, in tabular form as described in Appendix A (Questionnaires). When an analysis of variance is used to test for the significance of the variation in the scoring results, state the results of this analysis.

f. Furnish meteorological data for each test.

6.3.2 Test Project Planning

Record the following:

a. Detailed test schedule for each test.

b. Types and lengths of all cross-country test courses to be used.

c. The record forms and test logs to be employed.

d. Details of the safety release.

e. All safety precautions to be followed including applicable safety SOPs.

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f. Test facilities, equipment, accessories and their operating instructions available for use.

g. Arrangements concluded with supporting or participating agencies.

h. Month, day and hour of the start and finish of the short term and the long term (surveillance) storage periods.

i. Photographic facilities to be used.

6.3.3 Test Personnel Training and Familiarization

Record the following:

a. For all test personnel:

1) MOS.
2) Rank.
3) Unit.
4) Experience.
5) Previous training and skill level.

b. Adequacy of draft technical manual for supporting personnel training.

c. Requirements for training aids.

d. Difficulties encountered in completing training program.

6.3.4 Statistical Plan

Record the following:

a. Whether the "attributes" approach is to be used in determining probability of mission success or the statistical method used.

b. The number of identical "experiments" or missions (NxT) to be made.

c. The reference item to be used for comparison purposes.

d. The personnel to be used in the various tests.

6.3.5 User Test

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Record the following:

a. The test areas and operations used for evaluating the use of the test item.

b. A sketch map, or photomap, of the test area(s), and a description of the environment.

6.3.6 Initial Inspection

Record the following:

a. Conformance with specified packing, packaging, preserving, identification and marking criteria.

b. Any omissions of components or equipment, damage detected, and repairs made.

c. Physical characteristics of the test item as specified in the applicable Volume 8 commodity item MTP, and in MTP 8-3-500.

d. Missing or improper protective coatings on the test item, and photographs of roughness, corrosion or microbiological growths.

6.3.7 Initial Operation

Immediately after assembling and/or emplacing or positioning the test item in the physical environment in which it will be used, record the performance characteristics obtained in all specified modes and ranges of operations under the applicable meteorological conditions.

6.3.8 Operational Test

Record the following:

a. The performance characteristics described under paragraph 6.3.7 b. during the dry and rainy seasons for all of the "experiments" or missions performed.

b. The terrain conditions encountered, including water velocities and depths at stream or river crossings.

6.3.9 Short Term Storage

Record the following:

a. Condition of the test item(s) just prior to storage, and the storage preparations made, including (if applicable) the shipping

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containers used.

b. Storage environment and daily meteorological conditions.

c. Number of items involved.

d. Method and duration of storage of each item, and the calendar dates of all inspections.

e. Results of all inspections made.

f. Maintenance and repairs made.

g. Extent of deterioration of operating characteristics.

6.3.10 Surveillance

Record the following:

a. The condition of the test items just prior to storage, and the storage preparations made, including all containers and packaging used.

b. Storage conditions and daily meteorological conditions.

c. Number of items involved.

d. Method and duration of storage of each item and the calendar dates of all inspections.

e. Results of all inspections made.

f. Maintenance and repairs made.

g. Extent of deterioration of operating characteristics.

6.3.11 Maintenance Evaluation

Record the following:

a. Scheduled maintenance:

1) Description of test activity, environment, and meteorological factors associated with maintenance performed.
2) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.

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3) Component or feature involved.
4) Elapsed time since previous scheduled or unscheduled maintenance performed (whichever is shorter). Identity of previous maintenance performed (scheduled or unscheduled).
5) Parts required, if any.
6) Elapsed time required to complete maintenance actions.
7) Difficulties in completing maintenance actions.
8) Total scheduled maintenance time consumed in hours.

b. Unscheduled maintenance:

1) Description of test conduct, environment, and meteorological conditions under which maintenance requirements developed.
2) Procedures and special tools (if any) required to complete maintenance actions; description of maintenance performed.
3) Elapsed time since previous scheduled or unscheduled maintenance performed (whichever is shorter). Identity of previous maintenance performed.
4) Parts required, if any.
5) Difficulties in completing maintenance actions.
6) Components or feature involved and method used to determine it.
7) Damage caused to associated parts of the system by failure, if applicable.
8) Reason for maintenance required, if ascertainable.
9) Recommendations to prevent recurrence of maintenance requirements.
10) Elapsed time to identify and locate fault.
11) Elapsed time to rectify fault.
12) Total unscheduled maintenance times consumed in hours.

c. Time to complete related maintenance functions.

1) Supply delay time.
2) Administrative delay time.
3) Technical engineering investigations or inspections related to analysis of cause of failure or detection of suspected incipient failures.
4) Initial and final technical inspections.

d. For all maintenance performed:

1) The adequacy of the draft technical manual and maintenance information provided.

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2) The adequacy of the OEM tools and repair parts.
3) The adequacy of the safety instructions.

e. Total operating time in hours.

f. Identify each maintenance action as either a chargeable or nonchargeable failure in accordance with definitions and illustrations provided in USATECOM Regulation 750-15.

1) Total number of chargeable failures.
2) Total number of nonchargeable failures.

6.3.12 Safety

Record the appropriate data as required by the specific Volume 8 MTP for the test item, MTP 8-3-506, AMCR 385-224, AMCR 385-235, the maintenance package and the following:

a. A tabulation of all safety features and/or devices.

1) Type of feature/device.
2) Purpose of the feature/device.
3) Suitability of the feature/device.
4) Adequacy of the feature/device.
5) Proper operation of the feature/device.

b. Adequacy of warning plates, instructions and markings in content, clarity, sufficiency and location.

c. Any condition that did or might present a safety hazard including cause and corrective action required to alleviate the condition.

d. Presence and adequacy of fire fighting equipment and suitability of stowage and control location.

e. Suggestions to improve safety features, safety measures and/or precautions.

6.3.13 Human Factors

Complete the task/item checklists by rating the inclusion of each design consideration as satisfactory or unsatisfactory.

6.3.14 Security from Detection

Record the following:

a. Characteristics of the terrain in which observations

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were made.

b. Implements used to make observations (field glasses, aircraft, etc.).

c. Distances from which test item observations were made.

d. Time of day and sky conditions when observations were made.

e. Type(s) of material used.

6.3.15 Value Analysis

a. Record appropriate comments for each of the topics listed below:

1) Mission Capacity.
2) Simplicity.
3) State of the Art.
4) Standardization.
5) Materials and Methods of Construction.
6) Tolerances.

b. When making recommendations for changes in test item features or components, record the following:

1) The feature or component under consideration.
2) Recommended change(s).
3) Reason(s) for recommended change(s).

6.4 DATA REDUCTION

6.4.1 General

Summarize all data using tabulations and/or charts as appropriate. Analyze and compare the data collected against specific criteria stated in governing documents. Provide a narrative description of the degree of suitability of the test item for use in the humid tropics. If the item is not suitable for use in the tropics, provide a complete description (including test results) of why the item is so adjudged.

6.4.2 Questionnaires

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When a group of persons are asked to fill out a questionnaire giving their opinion regarding specified features in a particular test, and an analysis of variance is used to test for the significance of the variation in the scoring results, perform this analysis as specified in Appendix A.

6.4.3 Checklists

Group the answers to all checklist questions under the applicable headings, such as safety, accessibility, and human factors, and arrange the answers in the order of their relative importance.

6.4.4 Maintenance Evaluation

Calculate the maintenance indicators as follows. Reference is made to USATECOM Regulation 750-15 for information as to the meaning and use of these indicators in assessing system maintenance.

a. Mean Time Between Failures (MTBF)

$$\text{MTBF} = \frac{\text{Total operating time (in hours)}}{\text{Total number of chargeable system failures}}$$

b. Mean Time Between Maintenance (MTBM)

$$\text{MTBM} = \frac{\text{Total operating time (in hours)}}{\text{Total number of scheduled and unscheduled maintenance actions performed}}$$

c. Mean Active Maintenance Downtime ($\bar{M}$)

$$\bar{M} = \frac{\text{Total scheduled and unscheduled maintenance time (in hours)}}{\text{Total number of scheduled and unscheduled maintenance actions}}$$

d. Mean Time to Repair (MTTR)

$$\text{MTTR} = \frac{\text{Total unscheduled maintenance time expended on chargeable failures (in hours)}}{\text{Total number of chargeable system failures}}$$

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e. Inherent Availability (A_i)

$$A_i = \frac{MTBF}{MTBF + MTTR}$$

f. Achieved Availability (A_a)

$$A_a = \frac{MTBM}{MTBM + \overline{M}}$$

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APPENDIX A

Preparation and Use of Questionnaires

1. Continuously Reuseable Test Items

When testing continuously reuseable test items such as protective boots, portable riot control agent dispensers and sampling and analyzing kits, follow the procedures described in Appendix B, MTP 2-4-003, substituting the words "convey", "technicans" and "technician" for the words "drive, "test drivers" and "driver", respectively.

2. Single Use Test Items

When testing single use items such as grenades, chemical land mines and chemical agent rockets, follow the procedures described in Appendix B, MTP 4-4-003.

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APPENDIX B

Checklists

The following checklist questions are presented under two categories. The first category entitled "Tropic Environment Considerations", deals with those features which are emphasized in a tropic environment. The second category, entitled "General Considerations", deals with those features which are substantially independent of the kind of climate involved. Under this category, documents are referenced in which such general checklist questions can be found.

I. TROPIC ENVIRONMENT CONSIDERATIONS

a. Are adequate drain provided to drain off moisture inside closed-off areas, caused by condensation due to temperature changes when exposed to high humidity conditions?

b. Are provisions such as shielding, insulation or warning labels provided when the temperature of surfaces with which personnel may come in contact exceeds 46.1°C. (115°F.)?

c. Is adequate protection provided on the test item to protect personnel and equipment from overhanging limbs and lianas?

d. Are provisions made on the test item to protect the operator/crew from the effects of prolonged exposure of the test item to the direct rays of the sun?

e. Is a suitable medical kit provided for on-the-site treatment against poisonous plants, insects or animals?

f. Is hermetic sealing used to keep out moisture whenever possible?

g. Is contact between corrodible metal parts and anti-fungus-treated materials avoided whenever possible?

h. Are drain holes provided, and are chassis and racks channeled to prevent moisture traps?

i. Are reflecting surfaces used on equipment which must be maintained while exposed to the sun?

j. Are materials used to the greatest extent possible for their non-corrosive and non-hygroscopic characteristics, as well as their ability to resist fungus growth?

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k. Are materials used which inhibit fungus growth rather than materials which include a fungicide, or which have received surface fungistatic or fungicidal treatment?

l. Are metallic parts designed without recesses, cups or traps where liquids can accumulate, and are metals in contact chosen as close together as possible in the electromotive force series or adjacent groups?

m. Are exposed surfaces completely covered with some form of protective coating or surface plating?

n. Are corrosion inhibitors or preservative materials used where necessary or possible?

o. Are synthetic rubbers which are resistant to both microbiological and ozone deterioration used wherever possible?

p. Is the item properly identified as being a chemical item?

q. Are provisions made to determine the kind and extent of color fading encountered?

II. GENERAL CONSIDERATIONS

Consult the following documents for the indicated checklist subject:

a. MIL-HDBK-1472

Checklist Number	Subject	Pages
A-1	Access (External)	A3-33
A-2	Latches and Fasteners (External)	A3-34,35
A-3	Latches and Fasteners (Internal)	A3-35,36
A-4	Access (Internal)	A3-36,37
A-5	Packaging	A3-37,38
A-6	Units-Parts (Failed)	A3-38
A-7	Visual Displays	A3-39
A-8	Fault and Operation Indicators	A3-39,40,41

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Checklist Number	Subject	Pages
A-9	Test Points (Availability)	A3-41
A-10	Test Points (Identification)	A3-42
A-11	Labeling	A3-42,43
A-12	Adjustments	A3-43,44
A-13	Testing (In Circuit)	A3-44,45
A-14	Protective Devices	A3-45,46
A-15	Safety (Personnel)	A3-46,47
B-1	External Test Equipment	A3-47,48
B-2	Connectors	A3-48,49
B-3	Jigs or Fixtures	A3-49
B-4	Visual Contact (with other team members)	A3-50
B-5	Assistance (Operations Personnel)	A3-50,51
B-6	Assistance (Technical Personnel)	A3-51
B-7	Assistance (Supervisors or Contractor Personnel)	A3-51,52

b. AMCP 706-134

Table No.	Subject	Pages
5-2	Mechanical/Functional Checklist	5-3,4
9-11	Controls and Displays	9-29,30
12-2	Accessibility	12-10,11
13-4	Identification	13-9,10
14-1	Interchangeability	14-2
15-3	Safety	15-9,10,11

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Table No.	Subject	Pages
16-3	Servicing	16-9,10
21-1	Fasteners	21-12,13
23-2	Equipment Units	23-15,16,17
23-3	Cabling and Connectors	23-25,26
23-4	Test Points	23-30,31
23-5	Test Equipment	23-34,35

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APPENDIX C

Sample Scenario

Location: US Army Tropic Test Center, Fort Clayton, Canal Zone (79°57'30"W,9°21'30"N)

Exposure: Tropical Rain Forest

Simulated Combat Scenario: An infantry squad consisting of a squad leader and twelve riflemen are assigned the task of observing a trail from a fixed postion. The briefing has described the trail as a possible source of supply to a hostile force to the south and that a group so engaged is expected to pass during the next twenty-four hours. The squad is informed that the supply operation personnel are part of a tactical unit and has used toxic chemical agents effectively against opposing forces.

a. The squad leader insures that each member of the squad has a serviceable and properly adjusted protective mask as well as other items of individual equipment required for the period of the exercise. He also obtained the company TOE chemical agent detector kit for possible use in the event of attack.

b. The squad moves out from the platoon position and moves undetected to its predetermined position to observe the trail.

c. The squad leader assigns areas of responsibility for observation of the trail and fields of fire for all around protection of the squad position.

d. All squad members take up their positions and camouflage them to provide protection from observation.

e. The scout for the supply group is fired upon by a member of the squad which fixes the squad's position for the security force of the supply group.

f. The hostile security force used an HD agent (use agent simulant) on the squad.

g. All squad members don their protective masks and hold their positions.

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h. The squad leader, after assuring that the hostile force has left the area, uses the chemical agent detector kit to determine the type of agent that was used against his squad.

i. Upon completion of mission, squad returns to assume its position in the platoon's defensive position.

NOTE: The foregoing mission outline is offered as a typical operational situation to exercise the test items and user in a realistic mission environment. The mission tasks of either group should not be restricted to the use of items provided as examples.

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APPENDIX D

Chemical Munitions, Weapons and Equipment Covered by this MTP

The items covered by this MTP include the following:

Bomblets, chemical.
Grenades, hand or weapon launched, smoke/incendiary, or riot control.
Grenades, hand or weapon launched, colored marking.
Guided missiles, chemical agent.
Mines, land, chemical or flame.
Multiple sub-munitions system, riot control.
Projectiles and cartridges, chemical, smoke/incendiary.
Rockets, chemical agent.

Individual Protective Equipment

Boots, protective.
Breathing apparatus, self-contained air/oxygen supply.
Ensemble, protective, supplied air.
Gloves, protective
Hood, protective
Liners, protective.
Masks, protective.
Overgarment, protective, disposable.
Respirators.

Dissemination Devices

Dispenser, riot control agent, portable.
Dispenser, riot control agent, vehicle or helicoptor-mounted.
Generators, smoke, mechanical.
Generators, smoke, pot.
Flame throwers.

Decontamination Devices

Decontamination apparatus, portable.
Decontamination apparatus, power drive, vehicluar or skid mounted.
Decontaimination kit, individual, field.

Detection Devices

Alarms, chemical.
Chemical agent detector kit.
Sampling and analyzing kit
Field laboratories.

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